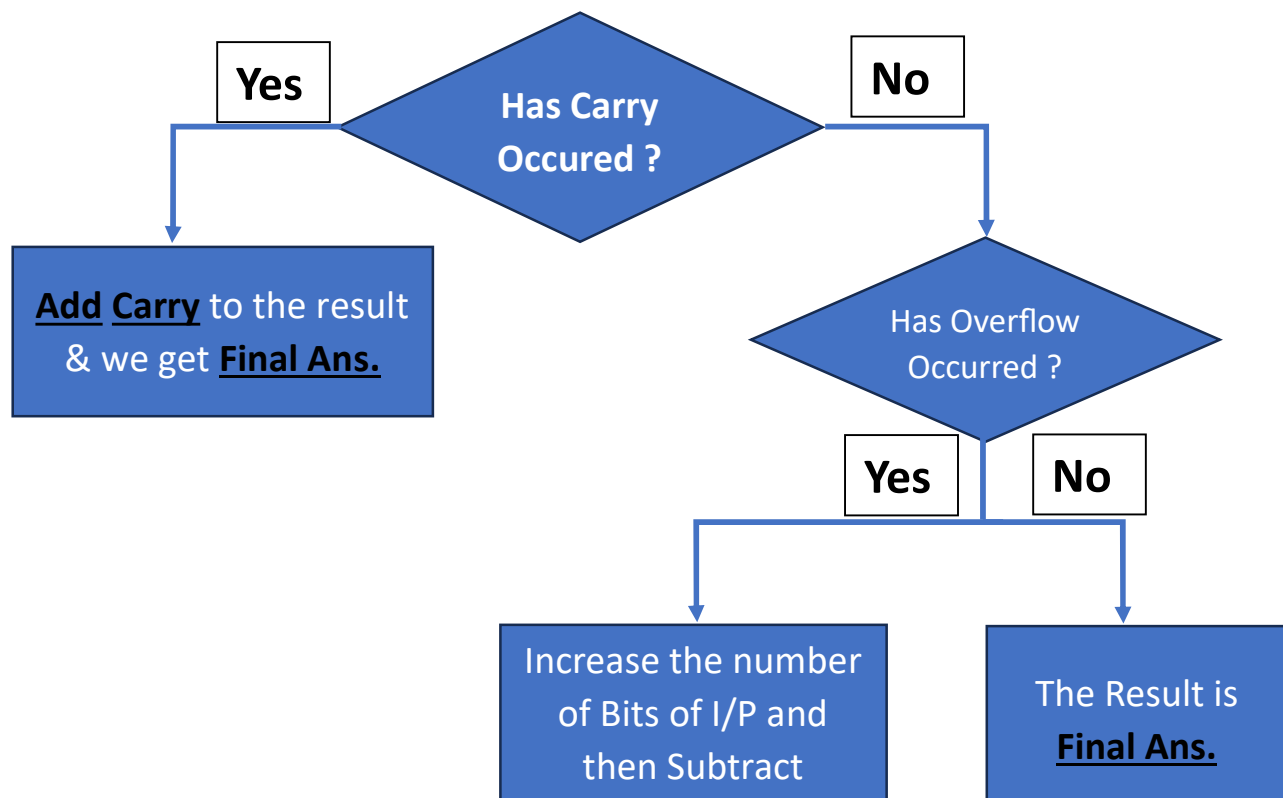


Subtraction using 1's Complement

- Step 1 : Take 1's complement of the subtrahend.
- Step 2 : Add two numbers.
- Step 3 :



1. How to Detect Overflow

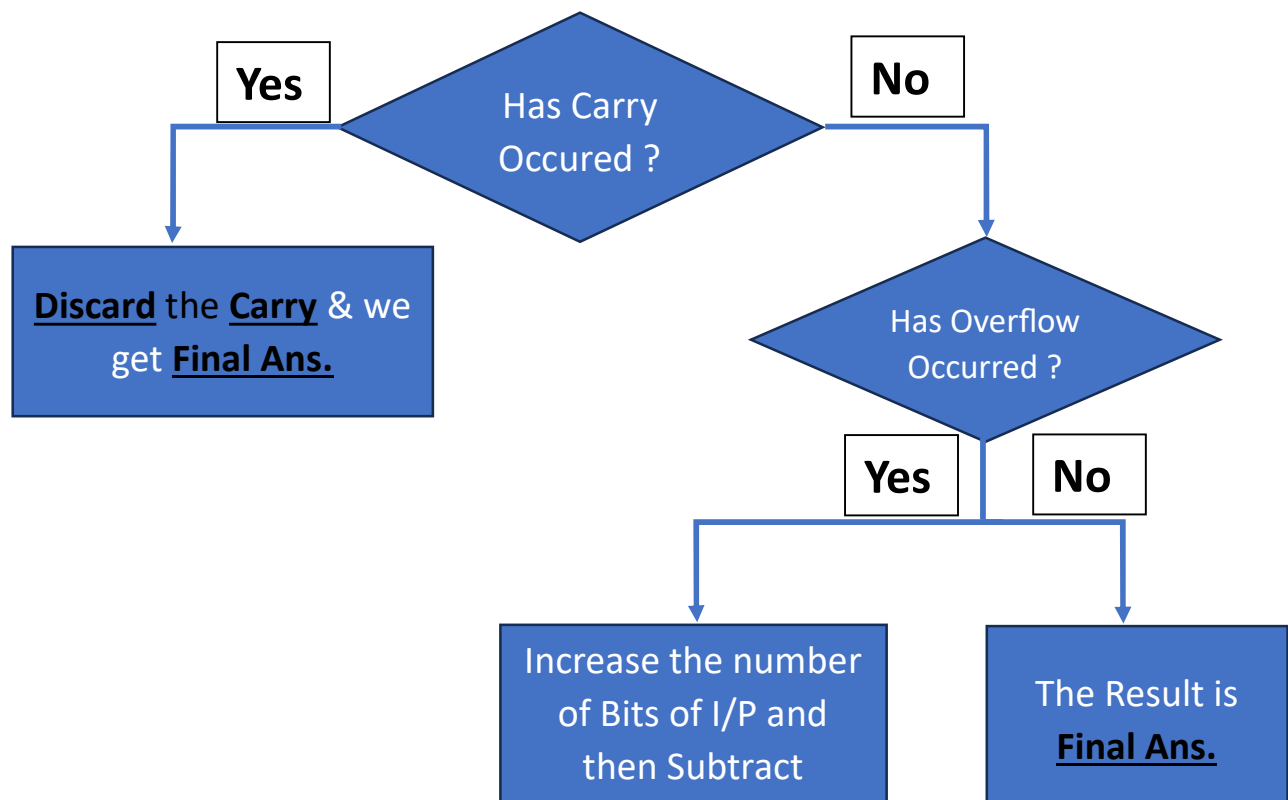
In 2's complement subtraction, you first convert the subtraction into addition (e.g., $A - B$ becomes $A + (-B)$). Overflow occurs if and only if: ⓘ

- You add two **positive** numbers and get a **negative** result.
- You add two **negative** numbers and get a **positive** result.

Note: If you add a positive and a negative number (like our $9 - 8$ example), **overflow is impossible**. The result will always fit within the original bit-width.

Subtraction using 2's Complement

- Step 1 : Take 2's complement of the subtrahend.
- Step 2 : Add two numbers.
- Step 3 :



12-bit 2's complement of -73.75 is

a. 01001001.1100

b. 11001001.1100

c. 10110110.0100

d. 10110110.1100

Posted under **Computer Architecture**

Answer: (c).10110110.0100

Subtraction of Number with larger bit-width

✚ To represent a number in a larger bit-width, you must fill the "extra" bits on the left with the value of the **Sign Bit** (MSB) :

- For **positive** numbers, the sign bit is 0, so you pad original number with 0s.
- For **negative** numbers, the sign bit is 1, so you pad with 1s after taking 2's complement.

❖ Addition of +9 and -8 using 12-bit 2's complement :

➤ +9 in 12-bit : 0000 0000 1001

➤ -8 in 12-bit :

○ +8 value in binary : 1000

○ +8 in 2's complement : 1000 (actually it is -8)

○ -8 after padding : 1111 1111 1000

➤ The whole Process looks like :

```
      0000 0000 1001  (+9)
+   1111 1111 1000  (-8)
-----
(1)0000 0000 0001  (Result)
```

<u>Q-2.</u> -73.75 represent in 12 bit 2's complement			
2	73	7	Integer fraction
2	36	0	
2	18	0	$0.75 \times 2 = 1.5$
2	9	7	$0.5 \times 2 = 1.0$
2	4	0	↓ 1 + 0.5
2	2	0	↓ 1 + 0.000
2	1	1	
	0		
$\therefore (73.75)_{10} = (1001001.11)_2$			
0110110.00 ← 1's			

- Add only one 1/0 as per the sign of the number to the **Right**.
- Take **2's complement** from the Fractional Rightmost value, keeping that 1st one added bit intact.
- So, the resultant will be 10110110.01
- Now **add 0s to the right for making** this number **12-bit**.
- So, the **Final ANSWER** will be : **10110110.0100**

Q $\Rightarrow$ Add 29 and -18 using 12-bit 2's complement

* Sol.ⁿ :-

$$29 \rightarrow (11101)_2$$

$$18 \rightarrow (10010)_2$$

- As, 29 is a +ve number, we will do padding with 0s

$$\therefore 29 \text{ in 12-bit} \rightarrow 000000011101$$

- As 18 is a -ve number, we will pad using 1s.

$$\therefore 29 + (-18)$$

$$-18 \text{ in 12-bit} : 11111110110$$

$$\text{Subtraction} : 29 + (-18)$$

	0	0	0	0	0	0	0	1	1	0	1	
	0	0	0	0	0	0	0	1	1	1	0	(+29)
+	1	1	1	1	1	1	1	0	1	1	0	(-18)
	1	0	0	0	0	0	0	0	1	0	1	

↓ Discard

+11 in Decimal

Ex. 1:- $(1100)_2 - (1010)_2$

step 1:- 1's complement of 1010
is 0101

step 2:-

$$\begin{array}{r} 1100 \\ + 0101 \\ \hline 10001 \\ + 1 \\ \hline 10010 \end{array} \rightarrow \text{Ans.}$$

Ex. 2:- $(1010)_2 - (1100)_2$

step 1:- 1's complement of 1100
is 0011

step 2:-

$$\begin{array}{r} 1010 \\ + 0011 \\ \hline 1101 \end{array}$$

↓ 1's complement

$$1-0010$$

↓

Ans. 0

★ 2's Complement Subtraction :-

Step 1:- Take 2's Complement of no. to be subtracted.

Step 2:- Add

Step 3:- If you get carry then discard it and if you don't get carry then again take 2's complement for ans.

In this case ans. will be -ve

$$\text{Ex 7 :- } (0111)_2 - (0010)_2 \quad \left| \begin{array}{l} 7 \rightarrow 0111 \\ 2 \rightarrow 0010 \end{array} \right.$$

$$\begin{array}{r} \cancel{0} \cancel{1} \cancel{1} \cancel{1} \\ \cancel{0} \cancel{0} \cancel{1} \cancel{0} \end{array}$$

Step 1:- 2's complement of 0010

$$\begin{array}{r} 11101 \\ + 7 \\ \hline 101110 \end{array}$$

Step 2:-

$$\begin{array}{r} 11 \\ 0111 \\ + 1110 \\ \hline \boxed{1}0107 \end{array}$$

Discard

$$\therefore \boxed{\text{Ans.} = 0101} = 4 \rightarrow \text{Dec.}$$

Ex. 2 :- $(0010)_2 - (0111)_2$

Step 1 :- 2's complement of 0111 is

$$\begin{array}{r} 1000 \\ + 0001 \\ \hline 1001 \end{array}$$

Step 2 :- $0010 + 1001$

$$\begin{array}{r} 0010 \\ + 1001 \\ \hline 1011 \end{array}$$

↓ 2's complement

$$\begin{array}{r} 0100 \\ + 0011 \\ \hline 0111 \end{array}$$

Ans.